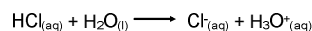


Strong vs Weak Acids

The strength of an acid refers to its ability to dissociate in solution.

Consider HCl:

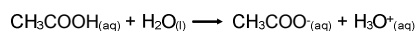


In this case, 100% of the HCl will dissociate in water. Therefore, the concentration of H_3O^{+} will be equal to that of the HCl.

Some strong acids:

HCl	hydrochloric acid
HNO_3	nitric acid
H_2SO_4	sulfuric acid
HBr	hydrobromic acid
HI	hydroiodic acid
HClO_4	perchloric acid

In the case of acetic acid:



the dissociation of ions is not complete (actually, only about 0.5%!); therefore the concentration of H_3O^{+} will not be proportional to the amount of CH_3COOH . This is considered a **weak** acid.

Other weak acids include HNO_2 , HCN, and HF

In a similar fashion, there are strong and weak bases:

Some strong bases:

LiOH	lithium hydroxide
NaOH	sodium hydroxide
KOH	potassium hydroxide
$\text{Ca}(\text{OH})_2$	calcium hydroxide
$\text{Ba}(\text{OH})_2$	barium hydroxide

Some weak bases:

include all amines:

NH_3 , CH_3NH_2 , $(\text{CH}_3)_2\text{NH}$

